

Spearman's Rank Order Correlation

- Correlation is a statistical technique used to measure the degree of relationship between two or more variables. When the variables are expressed in the form of ranks or ordinal data, the most commonly used correlation measure is **Spearman's Rank Order Correlation**.
- Spearman's Rank Correlation was developed by **Charles Edward Spearman (1904)** and is represented by the symbol r_s (**rho**).
- It measures the strength and direction of association between two ranked variables.
- Unlike Pearson's correlation coefficient, Spearman's correlation does not require normally distributed data and can be used with ordinal data.

“Spearman's Rank Order Correlation is a non-parametric statistical measure that determines the strength and direction of association between two sets of ranks.”

It is particularly useful when:

- Data are available in ranks.
- Data are ordinal in nature.
- The sample size is small.
- The assumptions of Pearson's correlation are violated.

Objectives of Spearman's Rank Correlation

Spearman's Rank Correlation is used to:

1. Measure the degree of agreement between two judges.
2. Compare rankings assigned by different evaluators.
3. Determine the relationship between two ordinal variables.
4. Analyze relationships when exact measurements are unavailable.
5. Study educational, business, psychological, and social science data.

Characteristics of Spearman's Rank Correlation

- It is a non-parametric statistical technique.
- It works on ranked data.
- It is represented by (r_s) .
- Its value ranges from -1 to +1.
- It measures monotonic relationships.

- It is less affected by extreme values (outliers).

Interpretation of Spearman's Correlation Coefficient

Value of r_s	Interpretation
+1.00	Perfect Positive Correlation
+0.75 to +0.99	Strong Positive Correlation
+0.50 to +0.74	Moderate Positive Correlation
+0.25 to +0.49	Weak Positive Correlation
0	No Correlation
-0.25 to -0.49	Weak Negative Correlation
-0.50 to -0.74	Moderate Negative Correlation
-0.75 to -0.99	Strong Negative Correlation
-1.00	Perfect Negative Correlation

Formula of Spearman's Rank Correlation

When there are no repeated ranks, Spearman's Rank Correlation Coefficient is calculated using the following formula:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

Symbol	Meaning
r_s	Spearman Rank Correlation Coefficient
d	Difference between corresponding ranks
d^2	Square of rank difference
$\sum d^2$	Sum of squared rank differences
n	Number of observations

Example 1: Correlation Between Rankings Assigned by Two Teachers

Problem Statement

Two teachers independently rank five students according to their classroom presentation performance.

The school administration wants to determine whether both teachers have similar opinions regarding student performance.

Data

Student	Teacher A Rank	Teacher B Rank
A	1	2
B	2	1
C	3	3
D	4	4
E	5	5

Step 1: Calculate Difference Between Ranks

$d = R_1 - R_2$

Calculation Table

Student	Rank by Teacher A (R ₁)	Rank by Teacher B (R ₂)	d	d ²
A	1	2	-1	1
B	2	1	1	1
C	3	3	0	0
D	4	4	0	0
E	5	5	0	0
Total				2

$$\left[\sum d^2 = 2 \right]$$
$$[n = 5]$$

Step 2: Apply Spearman's Formula

Substituting the values into the formula:

$$\left[r_s = 1 - \frac{6(2)}{5(5^2 - 1)} \right]$$

$$\left[r_s = 1 - \frac{12}{5(24)} \right]$$

$$\left[r_s = 1 - \frac{12}{120} \right]$$

$$[r_s = 1 - 0.10]$$

$$r_s = 0.90$$

Interpretation

The value of Spearman's correlation coefficient is:

$$r_s = 0.90$$

This indicates a **strong positive correlation**.

Conclusion

There is a high degree of agreement between the rankings assigned by Teacher A and Teacher B.

Example 2: Spearman Correlation with Repeated (Tied) Values

Problem Statement

A researcher wants to examine the relationship between student performance in Mathematics and Computer Science.

Some students have obtained identical marks in Mathematics.

Mathematics Marks

Student	Marks
A	90
B	80
C	80
D	70
E	60

Computer Science Rankings

Student	Rank
A	1
B	3
C	2
D	4
E	5

Assigning Ranks for Tied Values

Students B and C have identical scores.

The normal ranks would be:

2 and 3

Average Rank:

$$\frac{2 + 3}{2} = 2.5$$

Mathematics Rankings

Student	Rank
A	1
B	2.5
C	2.5
D	4
E	5

Calculation Table

Student	Mathematics Rank	CS Rank	d	d ²
A	1	1	0	0
B	2.5	3	-0.5	0.25
C	2.5	2	0.5	0.25
D	4	4	0	0
E	5	5	0	0
Total				0.5

$$\sum d^2 = 0.5$$

Apply Formula

$$r_s = 1 - \frac{6(0.5)}{5(5^2 - 1)}$$

$$r_s = 1 - \frac{3}{120}$$

$$r_s = 1 - 0.025$$

$$r_s = 0.975$$

Interpretation

$$r_s = 0.975$$

This indicates a **very strong positive correlation**.

Conclusion

Students who perform well in Mathematics tend to perform well in Computer Science.

Example 3: Correlation Among Three Subjects

Problem Statement

A college wants to determine whether student performance is consistent across three subjects:

- Mathematics
- Science
- English

Data

Student	Mathematics	Science	English
A	95	90	85
B	85	80	75
C	75	70	65
D	65	60	55
E	55	50	45

Important Note

Spearman's Rank Correlation compares only **two variables at a time**.

Therefore, three separate analyses are required:

1. Mathematics vs Science
2. Mathematics vs English
3. Science vs English

Analysis 1: Mathematics and Science

Rank Table

Student	Math Rank	Science Rank	d	d ²
A	1	1	0	0
B	2	2	0	0

C	3	3	0	0
D	4	4	0	0
E	5	5	0	0

$$\sum d^2 = 0$$

Applying the formula:

$$r_s = 1 - \frac{6(0)}{5(5^2 - 1)}$$

$$r_s = 1$$

Interpretation

Perfect positive correlation exists between Mathematics and Science.

Analysis 2: Mathematics and English

$$\sum d^2 = 0$$

$$r_s = 1$$

Interpretation

Perfect positive correlation exists between Mathematics and English.

Analysis 3: Science and English

$$\sum d^2 = 0$$

$$r_s = 1$$

Interpretation

Perfect positive correlation exists between Science and English.

Advantages of Spearman's Rank Correlation

1. Easy to calculate and interpret.
2. Suitable for ordinal data.
3. Does not require normal distribution.
4. Less sensitive to outliers.
5. Useful for small samples.

Limitations of Spearman's Rank Correlation

1. Applicable only to ranked or ordinal data.
2. Measures only monotonic relationships.
3. Correlation does not imply causation.
4. Tied ranks may require special treatment.

Problem 1: Student Mathematics and Science Rankings

Five students are ranked according to their Mathematics and Science performance.

Student	Math Rank (X)	Science Rank (Y)
A	1	2
B	2	1
C	3	3
D	4	5
E	5	4

Example 2: Employee Experience and Performance

A company wants to determine whether more experienced employees receive higher performance rankings.

Employee	Experience Rank	Performance Rank
A	1	1
B	2	3
C	3	2
D	4	4
E	5	5
F	6	6

Example 3: Negative Correlation

Students ranked in Mathematics and Sports.

Student	Math Rank	Sports Rank
A	1	5
B	2	4
C	3	3
D	4	2
E	5	1

Example 4: Repeated Values (Tied Ranks)

Teacher ratings and student satisfaction scores.

Teacher	Rating	Satisfaction
A	90	95
B	80	85
C	80	80
D	70	75
E	60	65

Example 5: Three Sets of Data

Student	Mathematics	Science	English
A	95	92	88
B	90	85	80
C	85	80	82
D	80	78	75
E	75	70	72